

GE | PDP | AB032 — Developer & QA Handoff

Project: Gear Elevation A/B test — Benefits panel variant **Experiment ID:** GE|PDP|AB032 **Deliverable:** `variant-ge-ab032.js` (single IIFE, paste into Varify.io) **Audience:** Engineers reviewing the code, QA analysts validating the variant before launch. **Source of truth for this doc:** `documentation.md` (full project history) and `briefing.md` (AB test hypothesis).

1. The idea, in one paragraph

PDPs on `gearelevation.com` lean on long, unstructured descriptions. Users skim, miss the value proposition, and bounce (~78% bounce rate). The hypothesis: surfacing **4 short, value-focused benefit bullets** near the top of each PDP — between the title/rating area and the price/Add-to-Cart box — will reduce cognitive load and lift add-to-cart and conversion. The bullets are pre-curated for 493 products and ship as static data inside the variant code.

This document covers what was built, why it was built that way, and how to QA it.

2. Success metrics (from `briefing.md`)

- ↑ Add-to-cart rate
- ↑ Final conversion rate
- ↓ Bounce rate

Targeting rule: pages whose URL contains `/products/`. Variant assignment and traffic split are handled by Varify, not by this code.

3. High-level architecture

```
Varify.io
  ↓ injects
variant-ge-ab032.js (IIFE)
  ├── Data:      BENEFITS_MAP (493 handles → 4 bullets)
  ├── Anchor:    .product-block--price (beforebegin)
  ├── Idempotent: #ge-ab032-benefits guard
  ├── SPA:       MutationObserver, throttled + capped
  ├── Tracking:  dataLayer.push(ge_ab032_variant_rendered)
  ├── Safety:    try/catch + console.warn, never breaks page
  ↓ renders
<div id="ge-ab032-benefits"> in the PDP info column
```

Key architectural choices and why:

Choice	Rationale
Inline <code>BENEFITS_MAP</code> (~125 KB) in the IIFE	No runtime fetch, no CDN failure mode, no flash-of-no-bullets. Strategy A from the spec.
Vanilla JS, no jQuery, no external assets	Snippet must be portable across themes and survive Shopify's script-defer pipeline.
BEM-scoped CSS under <code>#ge-ab032-benefits</code>	Avoids theme collisions without <code>!important</code> .
<code>MutationObserver</code> with 1/sec throttle, 10-render or 30s cap	Re-renders on SPA variant changes; can't run away.
<code>try/catch</code> at the boot boundary	Variant must never break the page, even if a Shopify theme update changes selectors.

4. The DOM anchor — the one thing that changed from the spec

The original spec (Section 3 of `documentation.md`) said to insert **after** `div.price.product__price`. On the live theme, that selector resolves *inside* the gray price/Popular-Add-Ons wrapper, which placed the panel inside the gray box rather than above it (visible regression vs. the Figma preview).

Final anchor priority (in `findAnchor()`):

1. `.product-block.product-block--price` with `beforebegin` — panel sits **above** the gray box as a direct child of `.product-details`.
2. `.product-block.product-block--title` with `afterend` — fallback when the price block is missing.
3. `h1.product-title` with `afterend` — last-resort fallback.

The PDP info column is built from `.product-block--*` siblings in this order:

<code>.product-block--title</code>	← product name, "few left" notice
<code>.product-block--@app</code>	← Loox rating widget
<code>.product-block--dynamic_list</code>	
<code>.product-block--price</code>	← gray box: price, Popular Add-Ons, color/qty, ATC
<code>.product-block--form</code>	← variant selectors, quantity, Add to Cart button

Inserting `beforebegin` of `--price` slots the panel between the rating area and the gray container.

5. Data: `BENEFITS_MAP`

- 493 entries; each value is a `string[4]`.
- Source: curated in earlier sessions from each PDP's actual content. See `documentation.md` SESSION 3/4 handoffs for the curation history.
- Quality bar (3–6 word, value-focused, no fluff, English-only) is locked. Do not mutate the bullets without re-reading the relevant PDP markdown in `pdp content/`.
- File size impact: ~125 KB compact JSON literal embedded as `var BENEFITS_MAP = {...};`.

Truncated-handle gotcha (important)

Shopify auto-truncates URL handles >120 chars. The live PDP URL may carry the **full** untruncated handle (e.g. `...-fk0301747018588026`) while `BENEFITS_MAP` keys use the **truncated/hashed** form (e.g. `...-f-07faac4e`).

`lookupBenefits()` handles this with a two-step lookup:

1. Exact match in `BENEFITS_MAP` → use it.
2. Take the first 80 characters of the page handle, scan `BENEFITS_MAP` keys for any that start with the same prefix → use that.
3. Otherwise bail silently.

Verified against a real truncated-handle product (`1pc-portable-camera-...`) in jsdom.

6. Handle resolution

`resolveHandle()` tries these sources in order and returns the first hit:

1. `window.ShopifyAnalytics.meta.product.handle` (most reliable, theme-set)
2. `window.location.pathname.match(/\/products\/([^\?#]+)/)[1]` (URL fallback)
3. `[data-product-handle]` element's dataset (last-resort fallback)

If none resolve, the IIFE logs `[GE-AB032] handle not found` and bails — better to show nothing than the wrong bullets.

7. Visual spec (from `style.css` tokens + Figma previews)

Token	Value
Font family	Assistant , system fallback
Bullet text size	18px desktop / 16px mobile
Bullet text color	<code>#000</code>
Bullet font weight	400
Line height	21px desktop / 20px mobile
Check icon	Inlined SVG, 20x20, dark slate circle <code>#354C5E</code> with white checkmark
Row gap (icon <code>?</code> text)	8px
Row gap (bullet <code>?</code> bullet)	12px
Container margin	12px 0 16px (top tightened from spec's 16px to hug the rating area)
Container padding	0 (no card background)

Responsive: flex column on all viewports, text wraps naturally, icon stays at the top via `align-items:center` .

8. Failure modes — what the IIFE refuses to do

The IIFE will bail (silently or with a `console.warn`) if any of:

- `location.pathname` doesn't include `/products/` .

- `#ge-ab032-benefits` already exists in the DOM (idempotency).
- `resolveHandle()` returns null.
- `lookupBenefits()` returns null (no exact match and no prefix match).
- `findAnchor()` returns null (theme changed all known selectors).
- Any uncaught exception inside `boot()` → caught by top-level try/catch, logged with `[GE-AB032]` error: prefix.

The IIFE will never throw a page-breaking error. This is a hard requirement — Varify variants run on production traffic.

9. Tracking

The IIFE pushes a single event to `window.dataLayer` per page load:

```
{ event: 'ge_ab032_variant_rendered', product_handle: '<resolved-handle>', bullet_count: 4 }
```

Pushed inside `render()` after the DOM injection succeeds. Skipped if rendering bails. Pushed once per handle change (re-render on SPA navigation re-pushes if the handle is different).

Varify will track variant exposure separately. This `dataLayer` push exists for funnel debugging in GTM/GA.

10. File inventory

File	Purpose	Touch this?
<code>variant-ge-ab032.js</code>	The IIFE — paste this into Varify	No, regenerate via build script
<code>benefits.json</code>	Curated bullets (source of truth)	Edit if bullets need revision
<code>documentation.md</code>	Full project history & session handoffs	Append handoff sections only
<code>briefing.md</code>	AB test hypothesis	Reference only
<code>local-page.html</code>	Saved PDP HTML used for selector verification	Reference only
<code>preview-desktop.png</code> / <code>preview-mobile.png</code>	Figma mockups	Reference only
<code>icon-check.svg</code>	Source of the inlined check icon	Reference only
<code>style.css</code>	Figma's generated CSS (token reference)	Reference only
<code>handoff-developers-qa.md</code> / <code>.pdf</code>	This document	Update if architecture changes

11. QA checklist — run all of these on a live PDP

QA should treat the IIFE as production code. Test on **at least 3 different products**: one with short bullets, one with the longest bullet in the catalog, one with a truncated-handle slug (any URL ending in `...-<8-hex-chars>`).

Functional checks

- ☐ Panel appears between the title/rating area and the gray price box (NOT inside the gray box).
- ☐ Exactly 4 bullets render.
- ☐ Bullets match `benefits.json` for the current product's handle.
- ☐ Check icon is visible left of each bullet, dark slate circle with white checkmark.
- ☐ No console errors.
- ☐ No layout shift on existing PDP elements.

Idempotency

- ☐ Paste the IIFE into the console a second time → panel does not duplicate.
- ☐ Hard refresh the page → panel still appears, no duplicate.

Responsive

- ☐ Resize browser to 375px width → text shrinks to 16px / 20px line-height, icons stay aligned, no horizontal scroll.
- ☐ No bullet wraps awkwardly. Test the longest bullet (e.g. `9-axis tracking captures every hit`).

Tracking

- ☐ After successful render, `window.dataLayer` contains an object with event: `'ge_ab032_variant_rendered'`, the resolved `product_handle`, and `bullet_count: 4`.

Failure-mode safety

- ☐ Load the IIFE on the homepage or a collection page → no panel injected, no console errors.
- ☐ Pick a product whose handle is NOT in `BENEFITS_MAP` → no panel, `console.warn` with `[GE-AB032] no benefits for handle: .`

Variant selection (SPA test)

- ☐ On a PDP with color/size variants, switch variants several times → panel stays in place, no duplicates, no flicker.

Truncated-handle product

- ☐ Open a PDP whose URL ends in `...-<8-hex-chars>` (e.g. one of the `feice-fashion-...-f-07faac4e` family) → panel still renders with the right bullets, even though the URL handle differs from the map key.

Visual regression

- ☐ Compare to `preview-desktop.png` and `preview-mobile.png`. Spacing within ~2px, color hex-exact (`#000` text, `#354C5E` icon background).

12. Open questions for the user — answered before final ship

These are defaults applied in the IIFE header comment block. **Confirm or override before launch.**

#	Question	Default	Where to change
1	Verify snippet size limit	Assume ~133 KB is accepted (Strategy A)	Switch to Strategy B (hosted JSON) if rejected
2	dataLayer event name	ge_ab032_variant_rendered	EVENT_NAME constant
3	Non-English locales	Show on all locales (English bullets)	No-op in code; add locale check to disable
4	Missing handle behavior	Silent bail + <code>console.warn</code>	<code>render()</code> early returns
5	Per-bullet click tracking	Not implemented	Out of scope for v1

13. Code review checklist (for the engineer reviewing the IIFE)

- ☐ All code wrapped in IIFE; no globals leak.
- ☐ `'use strict'` at top.
- ☐ No `!important` in CSS.
- ☐ All selectors namespaced under `#ge-ab032-benefits`.
- ☐ No `document.write`, no inline event handlers.
- ☐ Top-level `try/catch` wraps the boot path.
- ☐ `MutationObserver` is disconnected after `MAX_RENDERS` or `OBSERVER_MAX_MS`.
- ☐ No external network calls, no `<img>` or `<link>` to remote assets.
- ☐ SVG icon is inlined, not fetched.
- ☐ Element creation uses `document.createElement` + `textContent` for bullet text (no `innerHTML` for user-controlled data). `innerHTML` is used only for the static SVG constant.
- ☐ All `console.*` calls are wrapped in `try/catch` (some Shopify themes clobber console).

14. Rebuild process — for future maintenance

If `benefits.json` changes:

```
# Build script reads benefits.json + /tmp/ab32_iife_template.js, writes variant-ge-ab032.js
python3 /tmp/ab32_build.py
```

If the IIFE template changes (anchor, styles, tracking, etc.):

1. Edit `/tmp/ab32_iife_template.js` (the template uses `__BENEFITS_MAP__` as the data placeholder).
2. Re-run the build script above.
3. Re-run `node --check variant-ge-ab032.js` to confirm syntax.
4. Re-run the QA checklist above on a live PDP.

To rebuild this PDF:

```
pandoc handoff-developers-qa.md -s -o /tmp/handoff.html \
  --metadata title="GE PDP AB032 - Developer & QA Handoff" \
  --css=/dev/null
weasyprint /tmp/handoff.html handoff-developers-qa.pdf
```

15. Glossary

- **PDP** — Product Detail Page (/products/<handle> on Shopify).
- **IIFE** — Immediately-Invoked Function Expression. The variant code is wrapped as `(function() { ... })()`; so it can't leak globals.
- **Verify** — Third-party A/B testing tool. Handles traffic split and variant injection.
- **Handle** — Shopify's URL-safe product slug. Some are truncated/hashed when very long.
- **Anchor** — The DOM element the IIFE inserts the panel relative to.

*End of handoff. Questions: refer to `documentation.md` **SESSION 5 BUILD COMPLETE** for the latest project state.*